| MCS-204 | OBJECT-ORIENTED MODELING | L T P<br>3 1 0 | 4 Credits |
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**Course Objectives:**

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| **CO1** | To understand the Object-based view of Systems |
| **CO2** | To develop robust object-based models for Systems |
| **CO3** | To inculcate necessary skills to handle complexity in software design |
| **CO4** | To Understand MDA |
| **CO5** | To understand the concept of model-based testing |

**Module I**

Unified Modeling Language, (UML), Use case modeling, Methodologies for object-oriented analysis and design (OOAD),

**MODULE II**

Design patterns, CASE tool support for OOAD and automatic code generation, Precise modelling (using OCL-Object Constraint Language) and analysis of software models,

**MODULE III**

Model-driven architecture (MDA), Modeling language design: meta-modeling, UML Profiles Advanced modeling topics: Aspect oriented modeling, Modeling non functional properties, roundtrip engineering, model-based testing, open research questions.

**Books and References:**


*1. Timothy Lethbridge , Robert Laganiere , "Object-Oriented Software Engineering: Practical Software Development using UML and Java", Mcgraw-hill.*

*2. Lethbridge and Laganiere, "Object-oriented Software Engineering", McGraw-Hill.*

**Course Outcome:**

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|---|---|
| **CO1** | Ability to analyze and model software specifications. |
| **CO2** | Ability to abstract object-based views for generic software systems. |
| **CO3** | Ability to deliver robust software components. |
| **CO4** | Ability to perform roundtrip engineering |
| **CO5** | Ability to define MDA |